

GRADE 9 MATHEMATICS

Term Examination - I | Academic Year 2026-27 | 5 October 2026

Maximum marks: 80

Duration: 3 hours

General instructions

- This question paper has five sections: A, B, C, D and E.
- Section A contains 20 multiple-choice questions carrying 1 mark each.
- Section B contains 5 short-answer questions carrying 2 marks each.
- Section C contains 6 short-answer questions carrying 3 marks each.
- Section D contains 4 long-answer questions carrying 5 marks each.
- Section E contains 3 case-study questions carrying 4 marks each (subparts of 1, 1 and 2 marks).
- All questions are compulsory, with internal choices where indicated.

SECTION A

20 questions × 1 mark each

1. If $P(E) = 0$, then the event E is (1)
(a) Certain event (b) Impossible event
(c) Equally likely event (d) Likely event
2. The factorization of $x^2 + 5x + 6$ is (1)
(a) $(x + 1)(x + 6)$ (b) $(x - 2)(x - 3)$
(c) $(x + 5)(x + 1)$ (d) $(x + 2)(x + 3)$
3. Find the coordinates of the midpoint of the line segment whose endpoints are A(4, 3) and B(−6, 5). (1)
(a) (−1, 4) (b) (3, 6)
(c) (−2, 8) (d) (5, −1)
4. The decimal representation of a rational number cannot be (1)
(a) Terminating (b) Non-terminating repeating
(c) Non-terminating non-repeating (d) None of these
5. The area of a circle whose circumference is 44 cm is (1)
(a) 77 cm^2 (b) 154 cm^2
(c) 616 cm^2 (d) 308 cm^2
6. The common difference of the AP −3, −1, 1, 3, ... is (1)
(a) −2 (b) 1
(c) 2 (d) 3
7. The decimal form of $56/1000$ is (1)
(a) 5.6 (b) 0.56
(c) 0.056 (d) 0.0056
8. The degree of the polynomial $4x^3 - 3x^2 + 7$ is (1)
(a) 1 (b) 2
(c) 3 (d) 4
9. Which of the following is a polynomial in one variable? (1)
(a) $x + \sqrt{x} + 3$ (b) $x^2 + 3x^{-1}$
(c) $x^3 - 2x + 5$ (d) $\sqrt{x} + 2$

10. The n th term of an AP with first term a and common difference d is (1)
 (a) $a + nd$ (b) $a + (n - 1)d$
 (c) $a - (n - 1)d$ (d) an
11. The next term of the AP 2, 5, 8, 11, ... is (1)
 (a) 13 (b) 14
 (c) 15 (d) 16
12. A kite is at the position $(-2, 5)$ on a coordinate grid. This point lies in (1)
 (a) Quadrant I (b) Quadrant III
 (c) Quadrant II (d) Quadrant IV
13. Using a suitable identity, the value of $51^2 - 49^2$ is (1)
 (a) 100 (b) 200
 (c) 300 (d) 400
14. How many tangents can be drawn to a circle from a point outside the circle? (1)
 (a) 1 (b) 2
 (c) 3 (d) 0
15. The expansion of $(x - 5)^2$ is (1)
 (a) $x^2 - 25$ (b) $x^2 + 10x + 25$
 (c) $x^2 - 10x + 25$ (d) $x^2 + 25$
16. The quotient of $(-5/8) \div (3/4)$ is (1)
 (a) $-6/5$ (b) $4/7$
 (c) $3/4$ (d) $-5/6$
17. In a circle of radius 14 cm, an arc subtends an angle of 45° at the center. The area of the sector is (1)
 (a) 71 cm^2 (b) 76 cm^2
 (c) 77 cm^2 (d) 154 cm^2
18. A rectangular board has length $(x + 4)$ cm and width $(x - 4)$ cm. What is its area? (1)
 (a) $x^2 - 16$ (b) $x^2 + 16$
 (c) $x^2 - 8x + 16$ (d) $x^2 + 8x + 16$
19. Assertion (A): Two coins are tossed. The number of elements in the sample space is 4. (1)
 Reason (R): When a coin is tossed n times, the number of elements in its sample space is 2^n .
 (a) Both A and R are true, and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A
 (c) A is true, but R is false (d) A is false, but R is true
20. Assertion (A): The sequence 5, 10, 15, 20, ... is an arithmetic sequence. (1)
 Reason (R): The difference between consecutive terms of an arithmetic sequence is constant.
 (a) Both A and R are true, and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A
 (c) A is true, but R is false (d) A is false, but R is true

SECTION B

5 questions $\times$ 2 marks each

21. Find the distance between the points $(-4, 0)$ and $(0, 3)$. (2)
22. The sides of a triangular playground are in the ratio 3:4:5. The perimeter of the playground is 240 m. (2)
 Find the area of the playground.

OR

The length of the minute hand of a clock is 7 cm. Find the area swept by the minute hand in 25 minutes.

23. Express 0.454545... in p/q form. (2)
24. Check whether $x = 2$ is a zero of the polynomial $p(x) = 2x - 4$. (2)
25. Which term of the AP 3, 8, 13, 18, ... is 78? (2)
- OR
- Find the number of terms in the AP 10, 7, 4, ..., -62.

SECTION C

6 questions × 3 marks each

26. Draw $\triangle PQR$ with $PQ = 6$ cm, $\angle P = 65^\circ$, $\angle Q = 55^\circ$. Draw the circumcircle of $\triangle PQR$. Is the center inside or outside the triangle? (3)
27. Do the points A(2, 1), B(6, 1) and C(2, 4) form a right-angled triangle? Verify using the distance formula. (3)
- OR
- The vertices of a triangular park are A(0, 0), B(12, 0) and C(12, 5). Find the lengths of all three sides of the park and hence its perimeter.
28. Find the zero of the linear polynomial $p(x) = 7x + 21$ and verify your answer by substitution. (3)
29. Prove that $\sqrt{3}$ is an irrational number. (3)
30. Find the 15th term of the AP 7, 13, 19, 25, ... (3)
- OR
- Find the common difference and the 10th term of the AP whose first term is 4 and whose 5th term is 20.
31. Factorize $125x^3 + 75x^2 + 15x + 1$ using a suitable identity and hence find its value when $x = 1$. (3)

SECTION D

4 questions × 5 marks each

32. A rectangular garden is planned on a coordinate grid. The four corners of the garden are marked as A(2, -2), B(8, 4), C(5, 7) and D(-1, 1). Verify whether the four points form a rectangle. Give a reason for your conclusion. (5)
33. (i) Simplify $(p^2 - 25)/(p^2 + 2p - 35) \times (p^2 - 49)/(p^2 - 12p + 35)$, where the denominators are not zero. (5)
- (ii) Using a suitable identity, evaluate $(102)^2$.
- OR
- (i) Factorize $x^3 + 12x^2 + 48x + 64$ and hence write the edge of a cube whose volume is this expression.
- (ii) If $a + 1/a = 6$, find the value of $a^2 + 1/a^2$.
34. A gym charges a fixed admission fee together with a fixed monthly membership fee. The total amount y (in rupees) payable for x months is given by $y = 30x + 500$. (5)
- (a) Complete the table of values for $x = 0, 2, 4, 6$.
- (b) Plot the ordered pairs on graph paper and draw the straight line representing $y = 30x + 500$.
- (c) Using the graph or otherwise, find the total amount payable for 12 months.

35. The sum of the first "n" terms of an AP is given by $a_n = 9n - 1$. (5)

(a) Find the first term and common difference of the AP.

(b) Check whether 145 is a term of this AP.

(c) Find the sum of the 5th term and the 10th term.

OR

If the 3rd term of a geometric progression is 12 and the 6th term is 96, find the common ratio and first 6 terms of this GP.

SECTION E

3 case-study questions × 4 marks each

36. A school in Dubai is designing a circular garden with a radius of 14 m. A sector of the garden with a central angle of 90° will be used as a flower bed. Take $\pi = 22/7$. (4)

(a) [1 mark] Find the length of the arc of the flower-bed sector.

(b) [1 mark] Find the area of the sector used as the flower bed.

(c) [2 marks] Find the area of the remaining part of the garden.

OR

Find the length of the arc of the remaining part of the garden.

37. The transport in-charge of a school recorded the arrival time of the school bus on 200 working days. (4)
The record is given below:

Arrival of the bus	Early	On time	Late by 5 min	Late by more than 5 min
Number of days	30	110	40	20

(a) [1 mark] Find the probability that on a randomly chosen day the bus arrived on time.

(b) [1 mark] Find the probability that the bus arrived early.

(c) [2 marks] Find the probability that the bus was late on a randomly chosen day.

OR

Find the probability that the bus was not late by more than 5 minutes.

38. A circular park has a radius of 13 cm. A chord AB of the circle is 10 cm long. A perpendicular OM is drawn from the center O to the chord AB. Find the length of AM. (4)

(a) [1 mark] Find the length of AM, where M is the midpoint of chord AB.

(b) [1 mark] Using right-angled triangle OMA, calculate the distance OM of the chord from the center.

(c) [2 marks] If another chord of the same length (10 cm) is drawn on the opposite side of the center, what will be its distance from the center? Explain why.

OR

A second chord of length 10 cm is drawn on the opposite side of O. Find its distance from the center. State the property of circles that justifies your answer.

Digitally re-typeset from a six-page photographed exam. School and student identifying details, answers and handwritten markings have been excluded. Mathematical questions follow the source wording, including an apparent inconsistency in Question 35.